# Maschke's Complete Reducibility Theorem

# An End-to-End Meta-Math Walkthrough

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Abstract

An arbitrary projection need not respect the group action. Averaging its conjugates creates a projection that does. The report presents the target theorem, declared imports, a dependency graph, five-faculty node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

**RT-01 / Representation Theory and Lie Theory / Report v1**

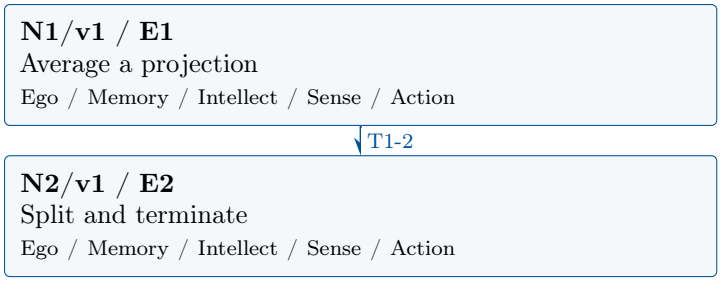
## I. TARGET AND IMPORT CONTRACT

**Theorem I.1.** *Let $G$ be a finite group, $k$ a field in which $|G|$ is invertible, and $V$ a finite-dimensional $k$-representation. Every invariant subspace $W$ has an invariant complement, and $V$ is a direct sum of irreducible representations. A representation is a homomorphism $\rho : G \to \mathrm{GL}(V)$; irreducible means nonzero with no proper nonzero invariant subspace.*

### A. Foundations and permitted imports

Use finite-dimensional basis extension to choose a linear projection onto a subspace, invariance under reindexing a finite group sum, and induction on dimension. The field may have positive characteristic provided it does not divide $|G|$.

## II. NODE INTERACTION WORKFLOW




Each box is one mathematical agent containing all five faculties. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

### A. Versioned evidence-transfer ledger

**T1-2: N1/E1/v1 to N2.** $P = |G|^{-1} \sum_g \rho(g) p \rho(g)^{-1}$ is equivariant, has image $W$, and restricts to the identity on $W$.

**Receiving activity.** N2.1

**Compatibility.** E1/v1 and finite $\dim V$. Check: The recursive decomposition terminates because each proper nonzero summand has smaller dimension.

## III. AGENT ROLES AND FIVE-FACULTY ACTIVITIES

### A. N1/v1: Average a projection

**Object and role.** Average $\rho(g)p\rho(g)^{-1}$ to enforce equivariance without losing the restriction to $W$. **Local obligation:** $P = |G|^{-1}\sum_g \rho(g)p\rho(g)^{-1}$ is equivariant, has image $W$, and restricts to the identity on $W$.

**Inputs.** An invariant $W \subset V$ and any linear projection $p : V \to W$ with $p|_W = \mathrm{id}$.

**E — Ego**

Publish E1/v1 only after establishing this obligation: $P = |G|^{-1}\sum_g \rho(g)p\rho(g)^{-1}$ is equivariant, has image $W$, and restricts to the identity on $W$.

**M — Memory**

An invariant $W \subset V$ and any linear projection $p : V \to W$ with $p|_W = \mathrm{id}$. Retain the stated assumptions and import versions.

**I — Intellect**

Average $\rho(g)p\rho(g)^{-1}$ to enforce equivariance without losing the restriction to $W$.

**S — Sense**

Division by $|G|$ must be legal, and each conjugate must still land in $W$.

**A — Action**

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N1.1 Choose a vector-space complement by extending a basis of $W$, and let $p$ be its projection. Each conjugate of $p$ lands in $W$, since $W$ is invariant under every $\rho(g)$. On $w \in W$, that conjugate sends $w$ to itself. Thus the average $P$ also maps into $W$ and is the identity there.

N1.2 For $h \in G$, conjugating $P$ by $\rho(h)$ replaces the index $g$ by $hg$, a permutation of $G$. Therefore $P$ commutes with $\rho(h)$. Its restriction to its image is the identity, so $P^2 = P$ and $\mathrm{im}\, P = W$.

**Export E1/v1.** $P = |G|^{-1}\sum_g \rho(g)p\rho(g)^{-1}$ is equivariant, has image $W$, and restricts to the identity on $W$.

**Acceptance gate.** Division by $|G|$ must be legal, and each conjugate must still land in $W$.

**Targeted repair.** If $p$ is assumed equivariant from basis extension alone, replace it by the conjugation average and verify the reindexing. Withhold this export and recheck its descendants after a versioned correction.

### B. N2/v1: Split and terminate

**Object and role.** Use the equivariant kernel as a complement, then induct on dimension. **Local obligation:** $V = W \oplus \ker P$ as representations, and every such $V$ is a finite direct sum of irreducibles.

**Inputs.** E1/v1 and finite $\dim V$. Incoming proof exports: E1/v1 at N2.1.

**E — Ego**

Publish E2/v1 only after establishing this obligation: $V = W \oplus \ker P$ as representations, and every such $V$ is a finite direct sum of irreducibles.

**M — Memory**

E1/v1 and finite $\dim V$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

**I — Intellect**

Use the equivariant kernel as a complement, then induct on dimension.

**S — Sense**

The recursive decomposition terminates because each proper nonzero summand has smaller dimension.

**A — Action**

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

**Numbered mathematical activities**

N2.1 Every $v$ decomposes as $Pv + (v - Pv)$, with the first term in $W$ and the second in $\ker P$. Their intersection is zero because $P|_W$ is the identity. If $u \in \ker P$, then $P\rho(g)u = \rho(g)Pu = 0$, so the complement is invariant.

N2.2 For $\dim V = 0$ the decomposition is the empty sum. A nonzero irreducible $V$ is already one summand. Otherwise choose a proper nonzero invariant $W$, split it as above, and apply the induction hypothesis to both summands of strictly smaller positive dimension. This yields a finite irreducible decomposition.

**Export E2/v1.** $V = W \oplus \ker P$ as representations, and every such $V$ is a finite direct sum of irreducibles.

**Acceptance gate.** The recursive decomposition terminates because each proper nonzero summand has smaller dimension.

**Targeted repair.** If repeated splitting is asserted to terminate without a measure, use the strictly decreasing dimensions of the summands. Withhold this export and recheck its descendants after a versioned correction.

## IV. CONTINUOUS MATHEMATICAL PROOF

### A. N1 — Average a projection

1. **N1.1.** Choose a vector-space complement by extending a basis of $W$, and let $p$ be its projection. Each conjugate of $p$ lands in $W$, since $W$ is invariant under every $\rho(g)$. On $w \in W$, that conjugate sends $w$ to itself. Thus the average $P$ also maps into $W$ and is the identity there.

2. **N1.2.** For $h \in G$, conjugating $P$ by $\rho(h)$ replaces the index $g$ by $hg$, a permutation of $G$. Therefore $P$ commutes with $\rho(h)$. Its restriction to its image is the identity, so $P^2 = P$ and $\operatorname{im} P = W$.

### B. N2 — Split and terminate

1. **N2.1.** Every $v$ decomposes as $Pv + (v - Pv)$, with the first term in $W$ and the second in $\ker P$. Their intersection is zero because $P|_W$ is the identity. If $u \in \ker P$, then $P\rho(g)u = \rho(g)Pu = 0$, so the complement is invariant.

2. **N2.2.** For $\dim V = 0$ the decomposition is the empty sum. A nonzero irreducible $V$ is already one summand. Otherwise choose a proper nonzero invariant $W$, split it as above, and apply the induction hypothesis to both summands of strictly smaller positive dimension. This yields a finite irreducible decomposition.

□

## V. WORKED ORCHESTRATION

Let $G = C_2$ act on $k^2$ by swapping coordinates, with $\operatorname{char} k \neq 2$. For $W = k(1,1)$, start with $p(x,y) = (x,x)$. Averaging its two conjugates gives $P(x,y) = ((x+y)/2, (x+y)/2)$. Its kernel is $k(1,-1)$, yielding the trivial and sign summands.

### A. Boundary and hypothesis audit

Over $\mathbb{F}_p$, let a generator of $C_p$ act by $T = I + N$ on a two-dimensional space, where $Ne_2 = e_1$ and $Ne_1 = 0$. Then $T^p = I$. The invariant line $ke_1$ has no invariant complement: any complementary line has a vector $e_2 + ae_1$, whose image differs by $e_1$ and cannot stay in that line. Here $|G| = 0$ in the field.

## VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

### A. Rejected argument, diagnosis, and correction

**Rejected N1 candidate:** the same average proves complete reducibility in every characteristic. **Sense:** the unipotent $C_p$ example contradicts the conclusion and $1/|G|$ is undefined. **Repair:** retain the stated invertibility assumption; no averaging repair exists for that counterexample under the broader claim. Recheck the complement and recursive decomposition with the correct field contract.

### B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Use invariant complements to decompose finite-group representations before character calculations; this does not itself prove the corresponding claim for arbitrary Lie algebras.

**Provenance.** This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

**Further reading.** Etingof and collaborators, Introduction to Representation Theory, sections on representations, characters, and $\mathfrak{sl}_2$; MIT Lie Groups and Lie Algebras I lecture notes for solvable Lie algebras.